



Republic of the
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IT123 CONFIGURING AND SECURING NETWORK SERVICES

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IT 123 – WEEK 7 LABORATORY

Lab Objectives

By the end of this lab, students will be able to:

1. Configure essential network services (DNS, DHCP, HTTP, FTP) in Ubuntu Server.
2. Verify service functionality using command-line tools.
3. Apply basic security measures (restricting access, enabling encryption where possible).
4. Understand the importance of service management in real-world IT infrastructure.

STUDENT EXERCISE

Scenario:

You are the system administrator of a small office network. You need to configure the server so that:

- Employees get their IP addresses automatically (DHCP).
- The company website (company.local) is hosted internally (DNS + Apache).
- File sharing via FTP is enabled only for local network users.

1. Configure company.local DNS zone

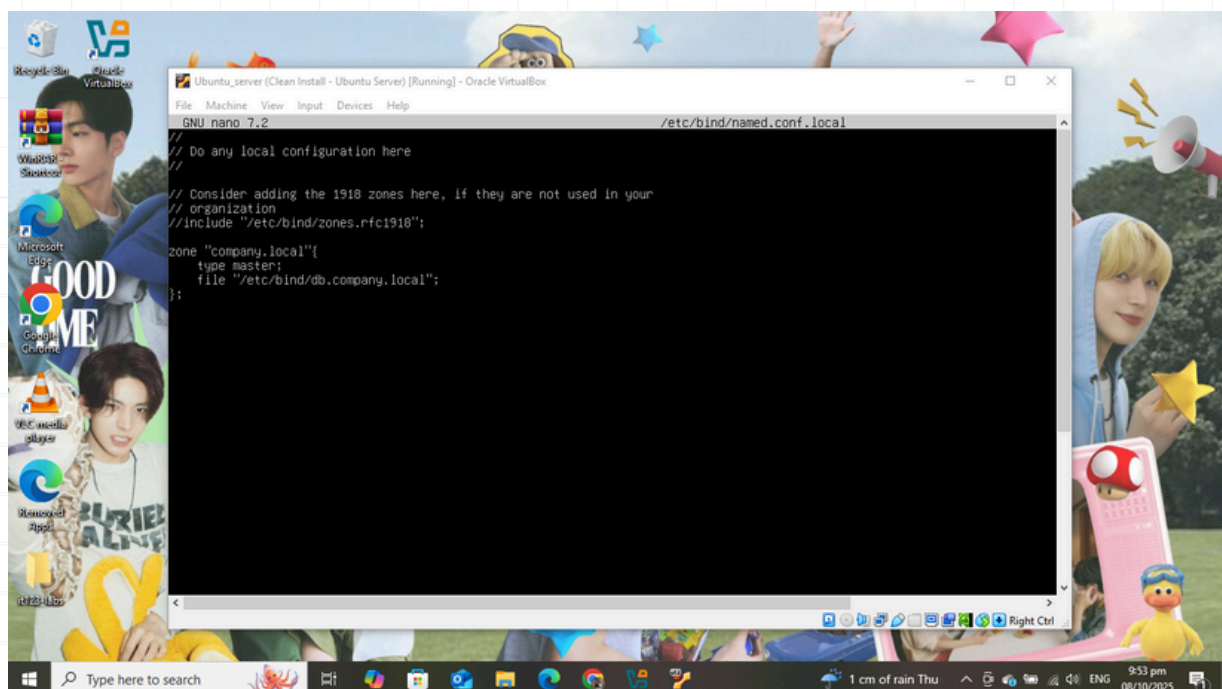
Step 1: Install BIND9

– `sudo apt install bind9 -y`

Step 2: Edit the DNS zone configuration

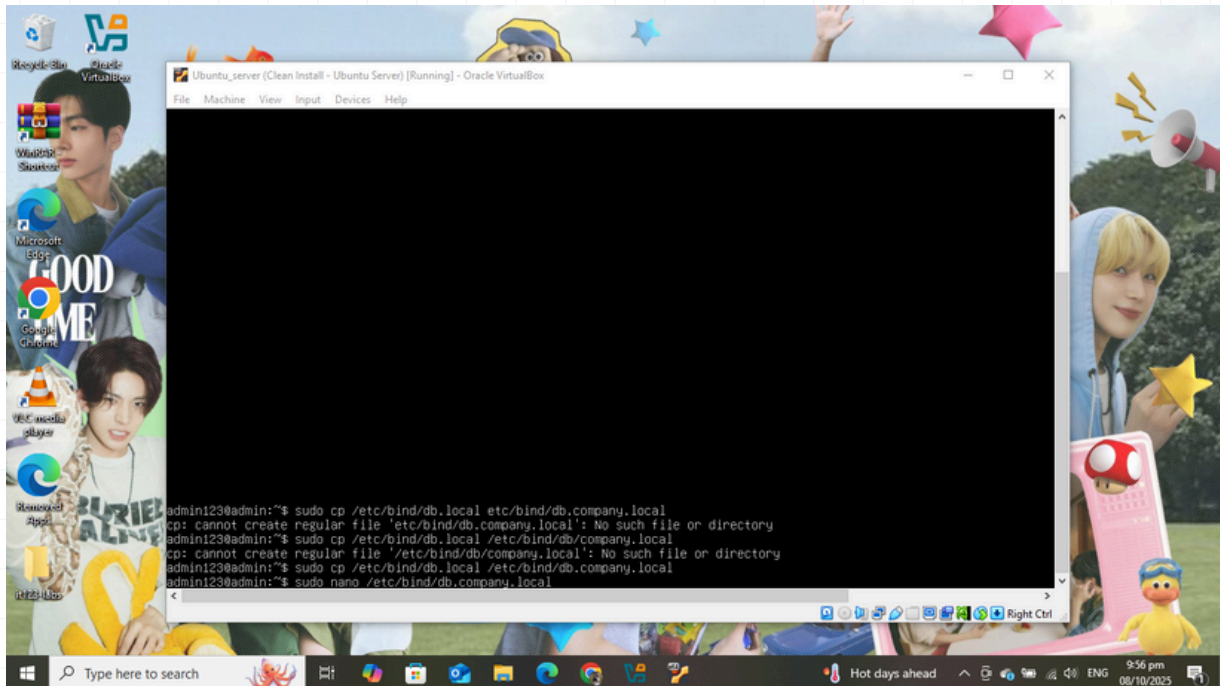
– `sudo nano /etc/bind/named.conf.local`

Add the following lines:



Step 3: Create the DNS zone file

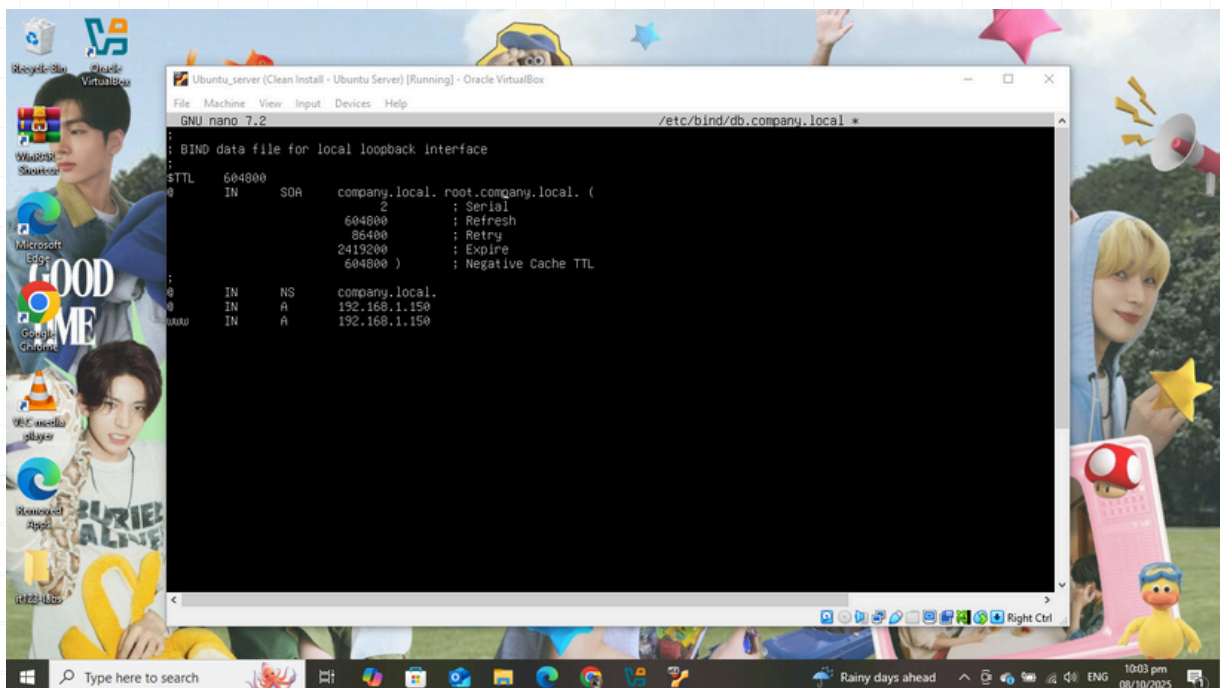
- `sudo cp /etc/bind/db.local /etc/bind/db.company.local`
- `sudo nano /etc/bind/db.company.local`



The screenshot shows a Windows 10 desktop with a colorful, anime-themed wallpaper. A VirtualBox window titled 'Ubuntu_server (Clean Install - Ubuntu Server) [Running] - Oracle VM VirtualBox' is open. Inside the window, a terminal shows the following commands and output:

```
admin123@admin:~$ sudo cp /etc/bind/db.local /etc/bind/db.company.local
cp: cannot create regular file '/etc/bind/db.company.local': No such file or directory
admin123@admin:~$ sudo cp /etc/bind/db.local /etc/bind/db/company.local
cp: cannot create regular file '/etc/bind/db/company.local': No such file or directory
admin123@admin:~$ sudo cp /etc/bind/db.local /etc/bind/db.company.local
admin123@admin:~$ sudo nano /etc/bind/db.company.local
```

Modify contents:



The screenshot shows the same VirtualBox window, but now the nano text editor is open, editing the file `/etc/bind/db.company.local`. The file contains the following DNS zone file configuration:

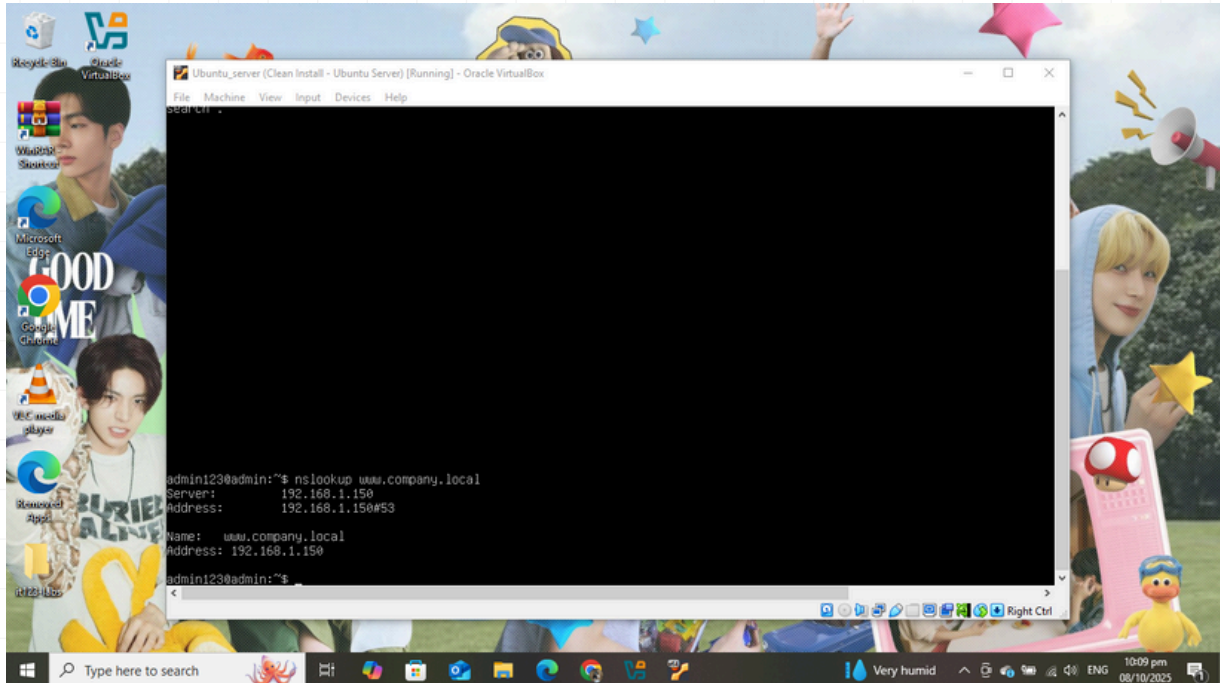
```
GNU nano 7.2 /etc/bind/db.company.local *
; BIND data file for local loopback interface
$TTL 604800
@ IN SOA company.local. root.company.local. (
    2      : Serial
    604800 : Refresh
    86400  : Retry
    241920 : Expire
    604800 ) : Negative Cache TTL
;
@ IN NS  company.local.
@ IN A   192.168.1.150
www IN A 192.168.1.150
```

Step 4: Restart BIND9

- `sudo systemctl restart bind9`

Step 5: Test DNS resolution

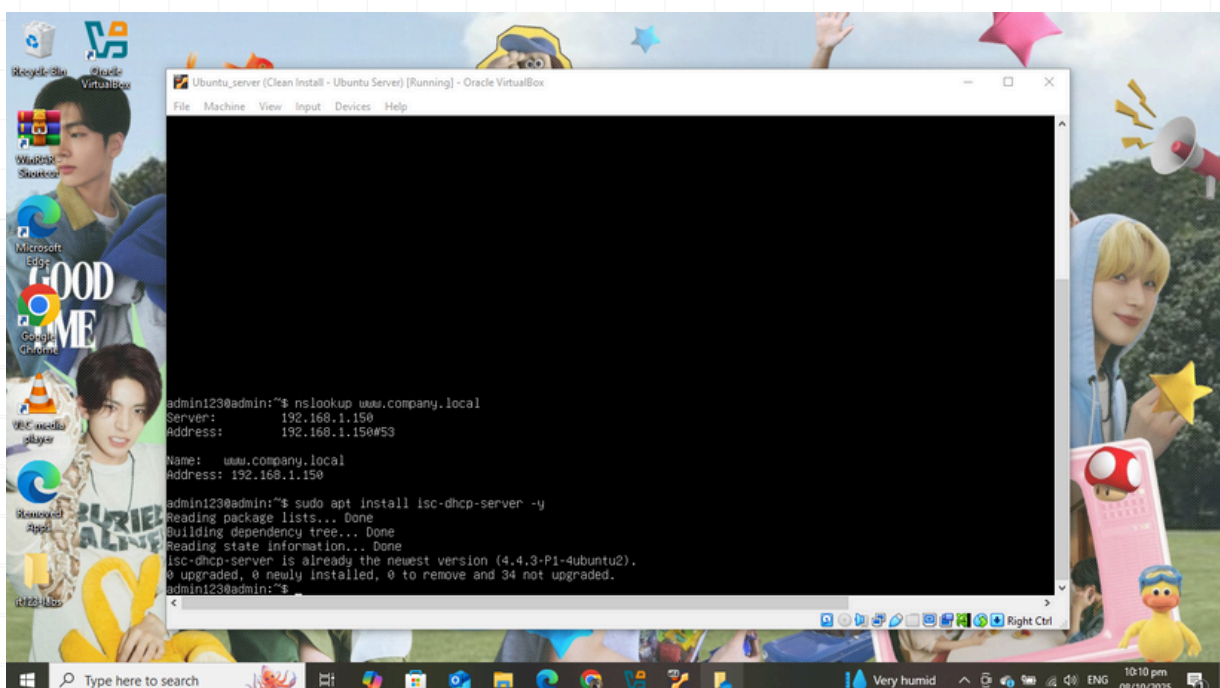
- `ping www.company.local`



2. Configure DHCP

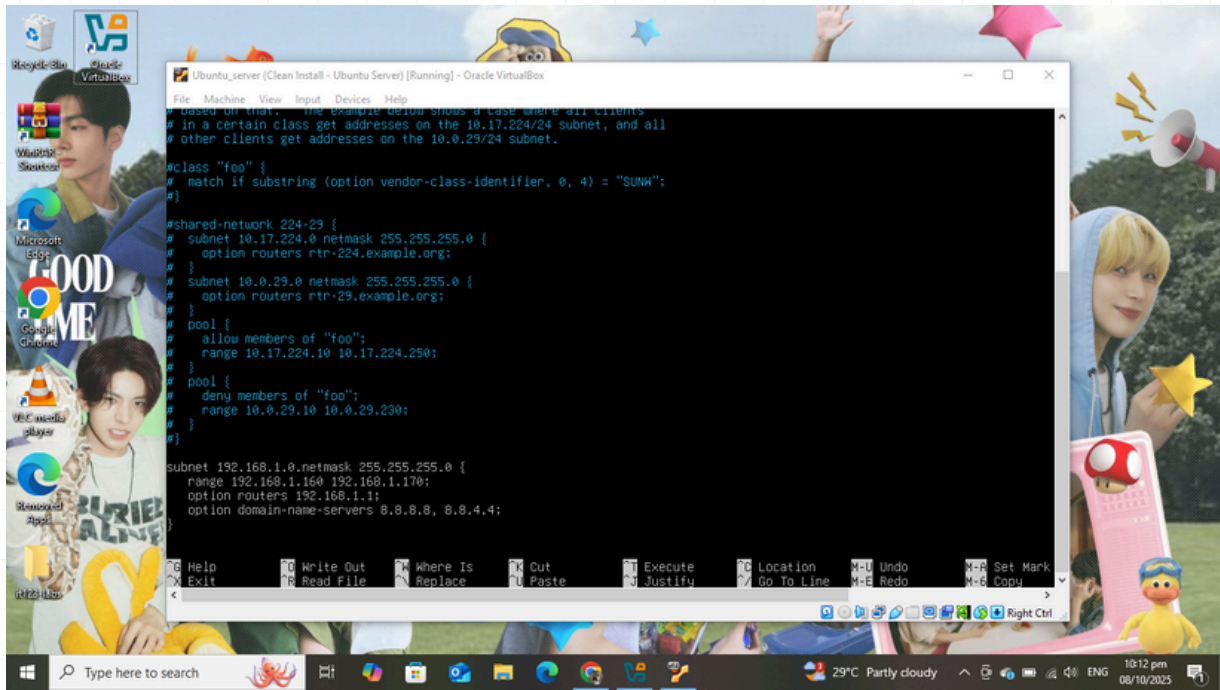
Step 1: Install DHCP server

- `sudo apt install isc-dhcp-server -y`



Step 2: Configure the DHCP range

- `sudo nano /etc/dhcp/dhcpd.conf`
- Add or edit lines:



```
File Machine View Input Devices Help
# A classless network where all clients
# in a certain class get addresses on the 10.17.224/24 subnet, and all
# other clients get addresses on the 10.0.29/24 subnet.

#class "foo" {
#  match if substring (option vendor-class-identifier, 0, 4) = "SUNW";
#}

#shared-network 224-29 {
#  subnet 10.17.224.0 netmask 255.255.255.0 {
#    option routers rtr-224.example.org;
#  }
#  subnet 10.0.29.0 netmask 255.255.255.0 {
#    option routers rtr-29.example.org;
#  }
#  pool {
#    allow members of "foo";
#    range 10.17.224.10 10.17.224.250;
#  }
#  pool {
#    deny members of "foo";
#    range 10.0.29.10 10.0.29.230;
#  }
#}

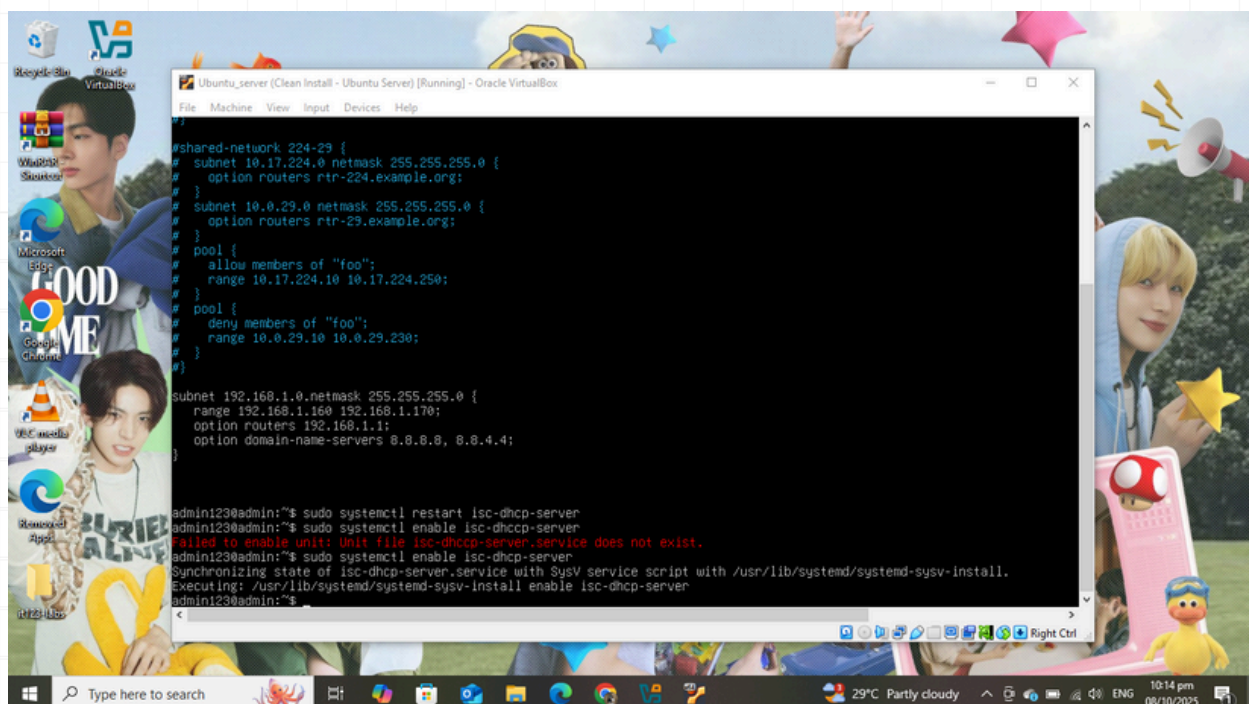
subnet 192.168.1.0 netmask 255.255.255.0 {
  range 192.168.1.160 192.168.1.170;
  option routers 192.168.1.1;
  option domain-name-servers 8.8.8.8, 8.8.4.4;
}
```

Step 3: Restart DHCP service

- `sudo systemctl restart isc-dhcp-server`

Step 4: Verify running status

- `sudo systemctl status isc-dhcp-server`



```
File Machine View Input Devices Help
#shared-network 224-29 {
#  subnet 10.17.224.0 netmask 255.255.255.0 {
#    option routers rtr-224.example.org;
#  }
#  subnet 10.0.29.0 netmask 255.255.255.0 {
#    option routers rtr-29.example.org;
#  }
#  pool {
#    allow members of "foo";
#    range 10.17.224.10 10.17.224.250;
#  }
#  pool {
#    deny members of "foo";
#    range 10.0.29.10 10.0.29.230;
#  }
#}

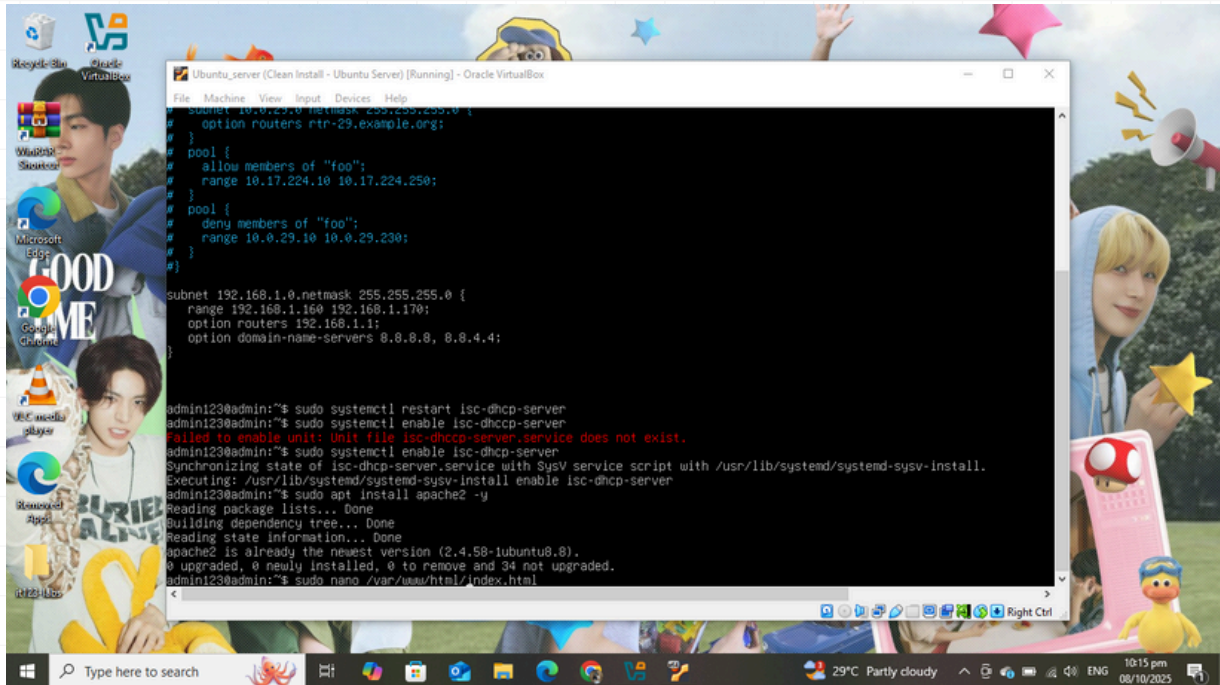
subnet 192.168.1.0 netmask 255.255.255.0 {
  range 192.168.1.160 192.168.1.170;
  option routers 192.168.1.1;
  option domain-name-servers 8.8.8.8, 8.8.4.4;
}

admin123@admin:~$ sudo systemctl restart isc-dhcp-server
admin123@admin:~$ sudo systemctl enable isc-dhcp-server
Failed to enable unit: Unit file isc-dhcp-server.service does not exist.
admin123@admin:~$ sudo systemctl enable isc-dhcp-server
Synchronizing state of isc-dhcp-server.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Execut ing: /usr/lib/systemd/systemd-sysv-install enable isc-dhcp-server
admin123@admin:~$
```

3. Configure Apache (Web Server)

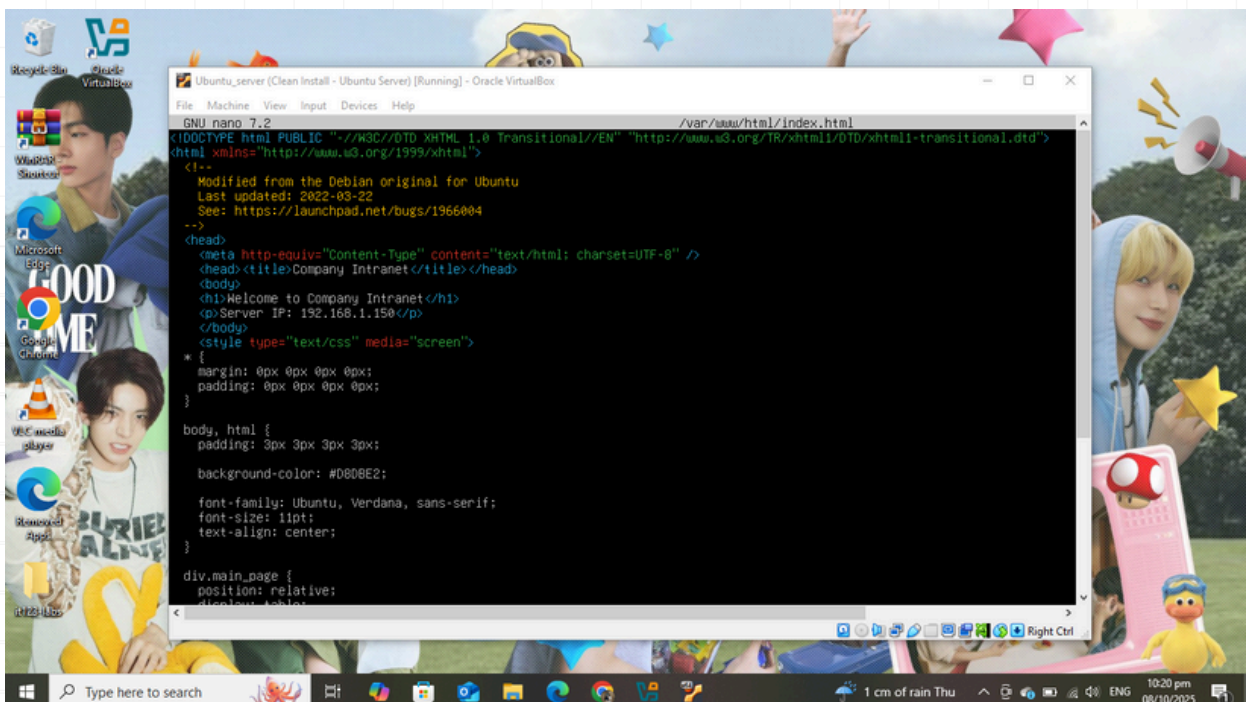
Step 1: Install Apache

- `sudo apt install apache2 -y`



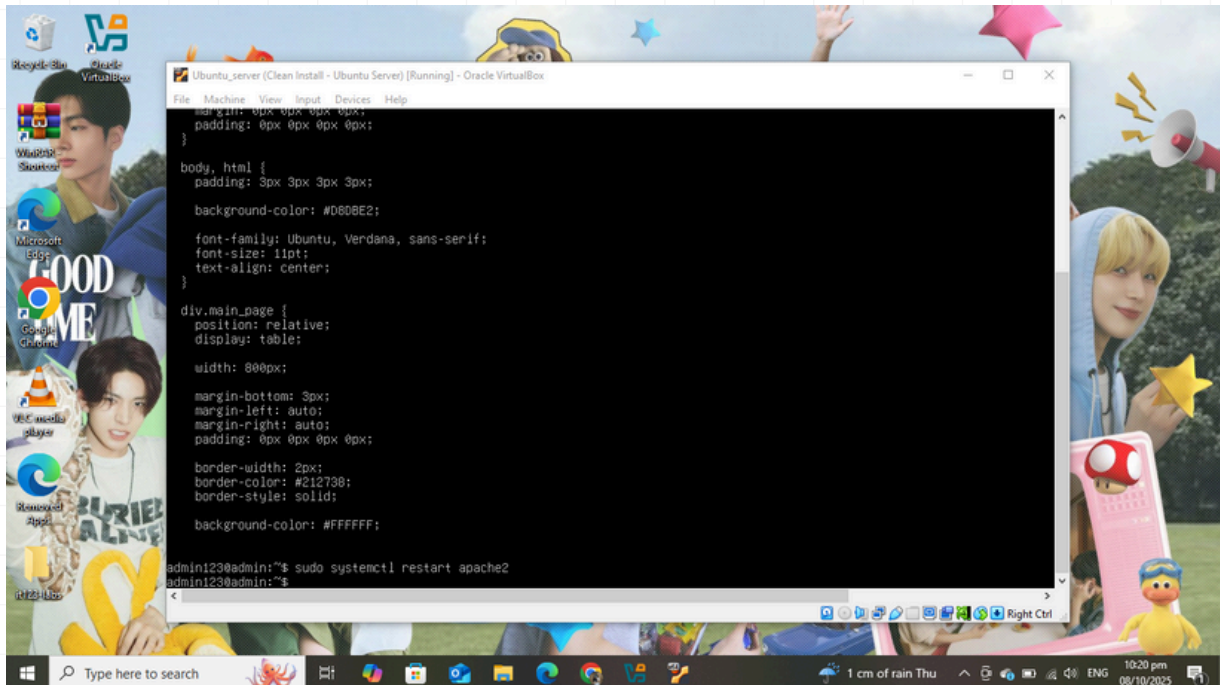
Step 2: Replace default web page

- `sudo nano /var/www/html/index.html`
- Replace content with:



Step 3: Restart Apache

- `sudo systemctl restart apache2`



Step 4: Test

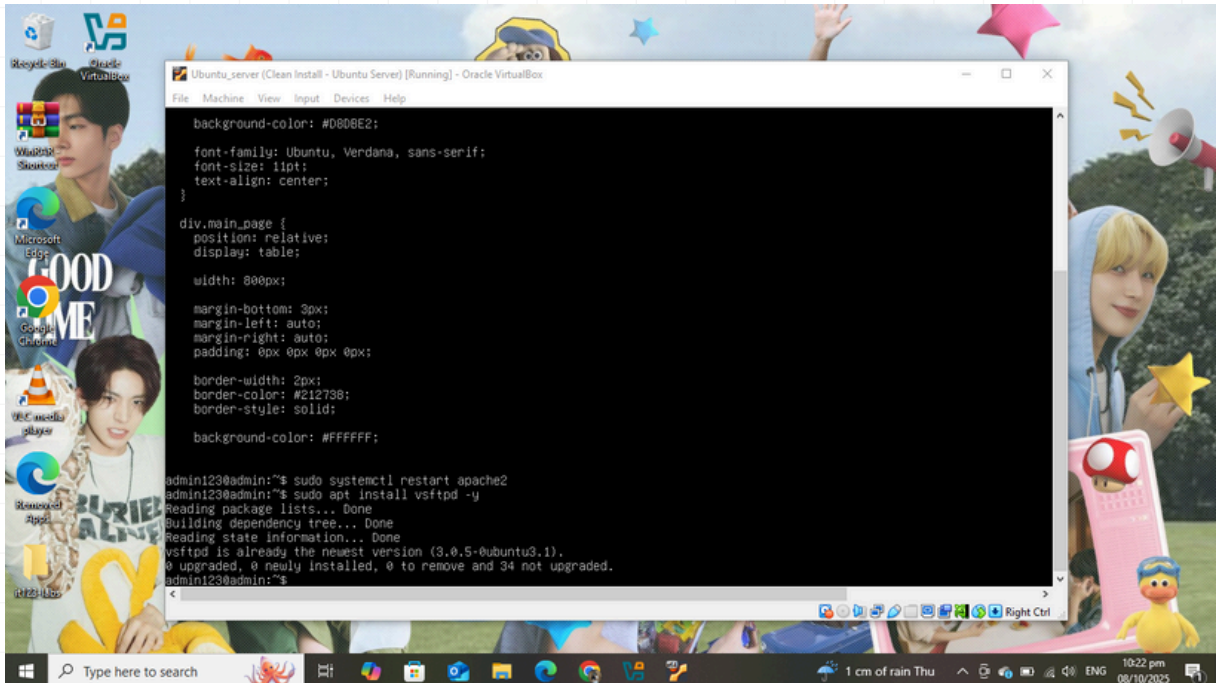
- Visit in browser:
<http://www.company.local>



4. Enable and Restrict FTP (Local Network Only)

Step 1: Install vsftpd

– sudo apt install vsftpd -y



Step 2: Configure FTP restriction

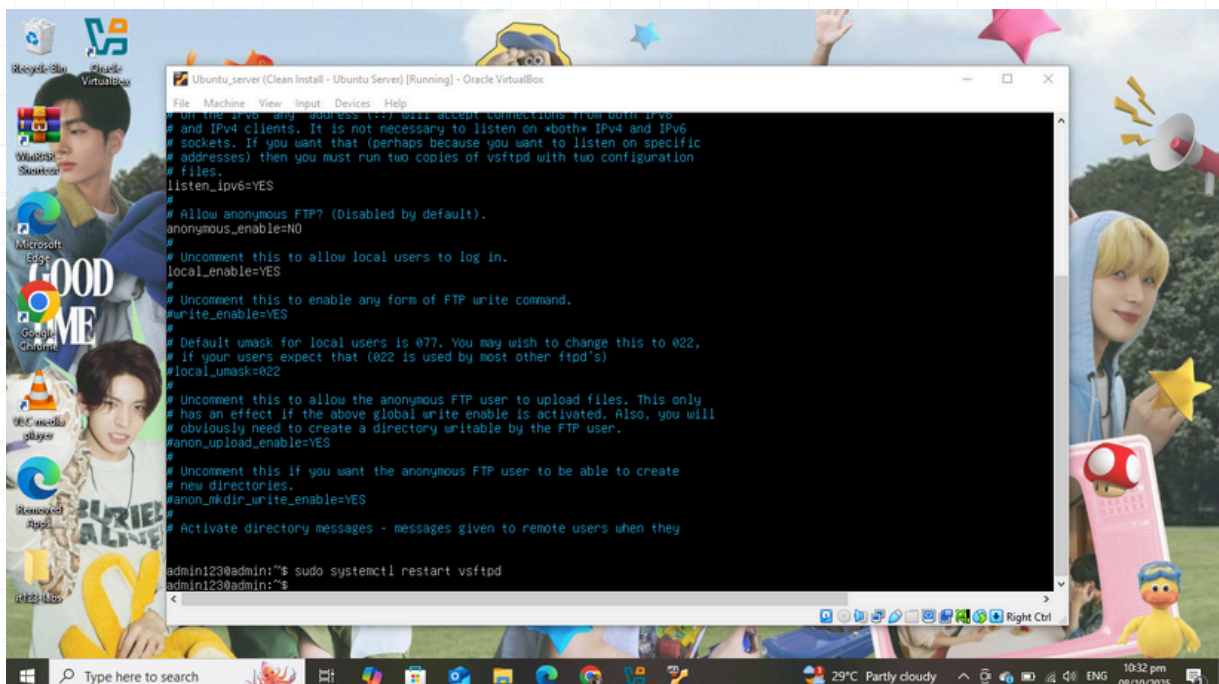
– sudo nano /etc/vsftpd.conf

– Add or ensure this line:

listen_address=192.168.1.150

Step 3: Restart FTP service

– sudo systemctl restart vsftpd



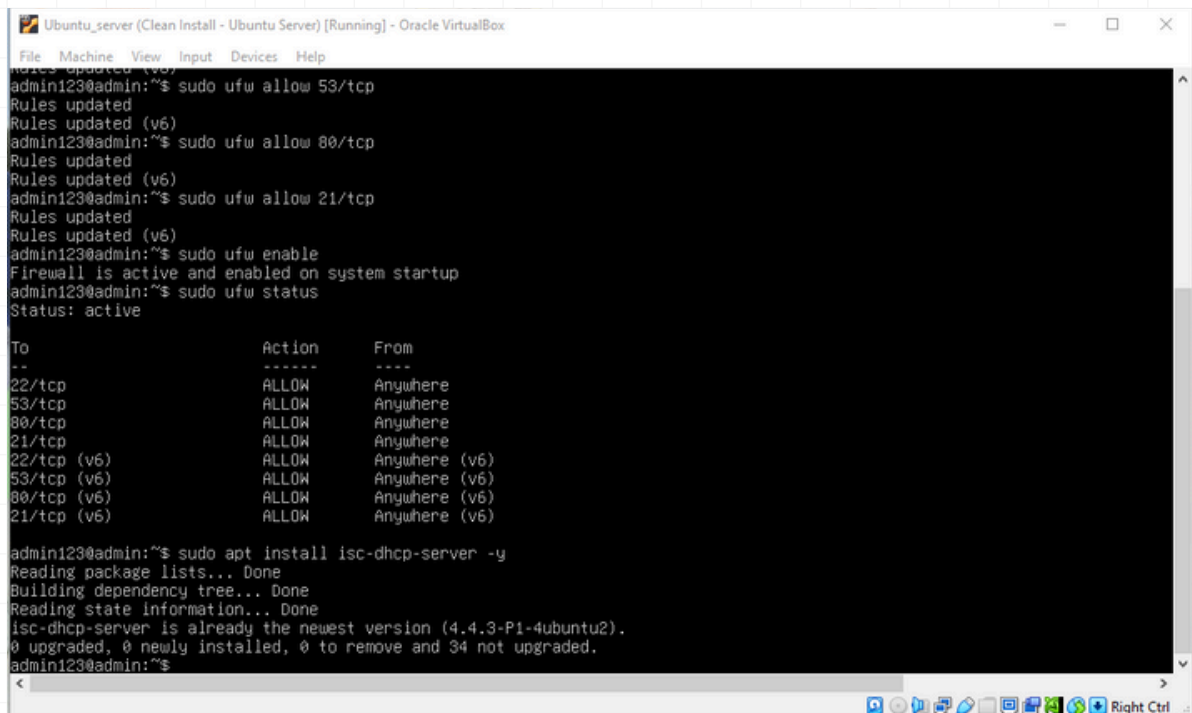
ADDITIONAL:

Step 1: Allow necessary ports

- sudo ufw allow 22/tcp
- sudo ufw allow 53/tcp
- sudo ufw allow 80/tcp
- sudo ufw allow 21/tcp
- sudo ufw enable

Step 2: Check status

- sudo ufw status



The screenshot shows a terminal window titled "Ubuntu_server (Clean Install - Ubuntu Server) [Running] - Oracle VirtualBox". The terminal output shows the following commands and their results:

```
admin123@admin:~$ sudo ufw allow 53/tcp
Rules updated
Rules updated (v6)
admin123@admin:~$ sudo ufw allow 80/tcp
Rules updated
Rules updated (v6)
admin123@admin:~$ sudo ufw allow 21/tcp
Rules updated
Rules updated (v6)
admin123@admin:~$ sudo ufw enable
Firewall is active and enabled on system startup
admin123@admin:~$ sudo ufw status
Status: active
```

To	Action	From
22/tcp	ALLOW	Anywhere
53/tcp	ALLOW	Anywhere
80/tcp	ALLOW	Anywhere
21/tcp	ALLOW	Anywhere
22/tcp (v6)	ALLOW	Anywhere (v6)
53/tcp (v6)	ALLOW	Anywhere (v6)
80/tcp (v6)	ALLOW	Anywhere (v6)
21/tcp (v6)	ALLOW	Anywhere (v6)

```
admin123@admin:~$ sudo apt install isc-dhcp-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
isc-dhcp-server is already the newest version (4.4.3-P1-4ubuntu2).
0 upgraded, 0 newly installed, 0 to remove and 34 not upgraded.
admin123@admin:~$
```